Project Proposal 3

Ultrasonic Measuring

Title: Wireless Distance Measurer with Smartphone Display and Logging

Application Description:
The Wireless Distance Measurer with Smartphone Display is a portable, Arduino-based project that allows users to accurately measure the distance to an object and view the results directly on their smartphone. Using an ultrasonic sensor, the system emits high-frequency sound waves and calculates the distance based on the echo return time. This data is transmitted wirelessly via Bluetooth to a custom mobile application. The app not only displays the real-time distance but also provides features like unit selection, data logging, and visual feedback. This project is ideal for a wide range of practical applications, from assisting with parking and checking clearance in tight spaces, to providing a simple tool for home improvement tasks, warehouse measurements, and robotic obstacle detection. It offers a modern, user-friendly alternative to traditional tape measures for quick and safe measurements.

Problem Identification:

Problem 1: Manual Tape Measures are Inconvenient and Inaccurate in Many Situations
Using a traditional tape measure can be difficult, especially when measuring long distances, hard-to-reach areas, or when a user is alone. It often requires two people, and human error in reading the markings can lead to inaccuracies. For parking, it’s impossible to use a tape measure to gauge the distance from a car bumper to a wall from the driver's seat.

Problem 2: Lack of Immediate Digital Data from Physical Measurements
When a measurement is taken, it is usually written down manually or remembered. This is inefficient and prone to being lost or forgotten. There is no easy way to digitally log measurements for later reference, track changes over time, or share them with others without re-measuring.

Problem 3: Difficulty in Measuring in Hazardous or Inaccessible Environments
Reaching out with a tape measure to check the height of a high ceiling, the depth of a pipe, or the distance to a moving object can be unsafe. A contactless measuring device allows users to stay at a safe distance while still obtaining accurate and reliable data.

Proposed Solutions:

Solutions for Problem 1: Manual Tape Measures are Inconvenient and Inaccurate

- Utilize an HC-SR04 ultrasonic sensor for contactless measurement, providing accurate distance readings.
- Display the reading in real-time on a smartphone, which the user can view from a distance.
- Include a feature to switch between metric (centimeters) and imperial (inches) units, making the data easier to interpret.
- Provide a visual or haptic (vibration/buzzer) alert on the mobile app when an object is within a user-defined "danger" or "target" zone.

Solutions for Problem 2: Lack of Immediate Digital Data from Physical Measurements

- Design the mobile app to log each measurement with a timestamp to create a history of readings.
- Allow users to add a simple text note to a specific reading (e.g., "Shelf Width," "Distance to Wall").
- Implement a "Share/Export" feature in the app to send the current measurement or a log of measurements via email or text message.

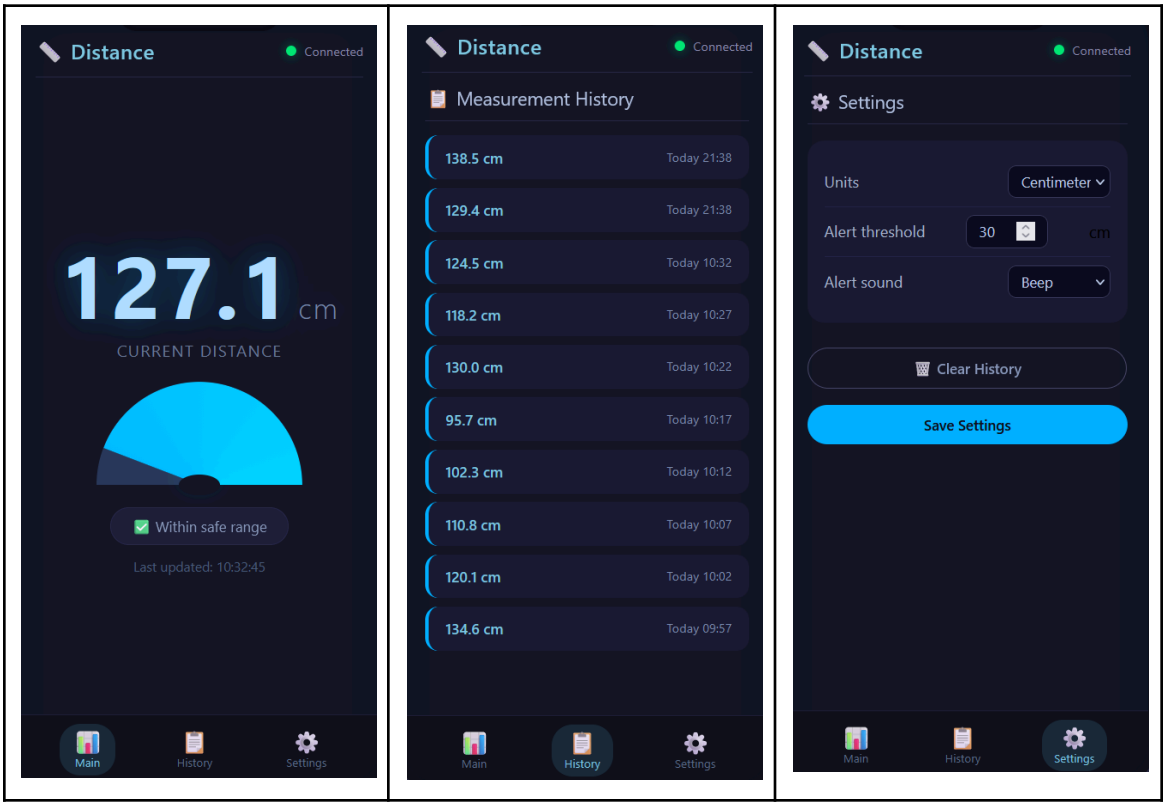
Solutions for Problem 3: Difficulty in Measuring in Hazardous or Inaccessible Environments

- Allow the user to trigger the measurement remotely from the smartphone app, eliminating the need to be near the Arduino to press a physical button.
- Enable continuous measurement mode, where the Arduino sends a stream of data to the app, allowing the user to see real-time changes as the sensor or object moves.
- Integrate the sensor into a small, portable, battery-powered enclosure, allowing it to be placed securely in a dangerous location while the user observes the readings from a safe distance.

Features and Functionalities:

- Real-Time Distance Display: The mobile app displays the measured distance in centimeters or inches as soon as the sensor captures the data.
- Wireless Data Transmission: The HC-05 Bluetooth module communicates the distance from the Arduino to the user's smartphone.
- Unit Selection: Users can easily switch between metric (cm) and imperial (inches) units within the mobile app for their convenience.
- Measurement Logging: The app saves the date, time, and measured distance of each measurement, creating a simple history log for the user to review.
- Visual Feedback on App: The app can display a simple progress bar, color-coded distance (e.g., green for safe, red for close), or numerical value for easy reading.
- Proximity Alert: The user can set a minimum distance in the app. If the ultrasonic sensor measures a distance below this threshold, an alert is triggered on the phone (and optionally, a buzzer connected to the Arduino).
- Record Keeping: Users can review past measurement logs to compare readings or track project progress.
- Clear and Reset Functionality: The mobile app includes a button to clear all logs and a reset function to re-initialize the sensor if needed.

User Interface


Distance
Connected
127.1 cm
CURRENT DISTANCE
Within safe range
Last updated: 10:32:45
Main
History
Settings
Distance
Connected
Measurement History
138.5 cm Today 21:38
129.4 cm Today 21:38
124.5 cm Today 10:32
118.2 cm Today 10:27
130.0 cm Today 10:22
95.7 cm Today 10:17
102.3 cm Today 10:12
110.8 cm Today 10:07
120.1 cm Today 10:02
134.6 cm Today 09:57
Main
History
Settings
Distance
Connected
Settings
Units
Centimeter
Alert threshold
30
cm
Alert sound
Beep
Clear History
Save Settings
Main
History
Settings